

CSCI 6450 – Project 2

IP Based IDS Simulation in Java

Due: 11:59 PM on 09/29/2025

Objectives

- Understand how IP based attacks work conceptually.
- Simulate IP based IDS.

Background

- An IP-based Intrusion Detection System (IDS) is a type of network security tool that focuses on monitoring and analyzing IP addresses in network traffic to detect suspicious or malicious activity. In this project, the simulated IDS inspects incoming IP addresses.

Assignment Requirements

1. Create Java Program(s)

- Write a single Java class for your project.
- Implement a simulation of IP based IDS.
- Note: you also need to create an ips.txt file which contains malicious IP



ips.txt

addresses. Here is an example.

2. Demonstrate the IDS

- Use a string of array to simulate network traffic, such as

```
String[] traffic = {  
    "192.168.1.100 → 8.8.8.8",  
    "10.0.0.10 → 1.1.1.1",  
    ...  
};
```

- Output should clearly show:
 - If a packet does not have a malicious IP, show it is ok.
 - If a packet has a malicious IP, show alert.

Please submit:

- **Source code (.java)** file(s).
- **The ips.txt created by yourself, which has at least 6 malicious IP addresses.**
- **Lab report:** Provide a short write-up (1–2 paragraphs) explaining:
 - Word document or pdf document.
 - What IP based IDS is.
 - Screenshots: Console output of tests. Please refer to the example question of signature based IDS.
 - If your running results meet all requirements, please state “my project meets the requirements”. Otherwise please point out the limitations in your project.
- **Please submit two individual files. Do not compress the two as a zip file.**

Grading Rubric (100 points)

Criterion	Points
Program correctness	
- Your program runs without syntax errors	20
- Your program can read the malicious IP addresses from the txt file	15
- Your program can detect malicious IPs	20
Code quality	
- Clear variable names, good structure, necessary comments	15
Lab report	
- Explanation of IP based IDS	15

Criterion**Points**

- Console output clearly shows ok or alert for normal or malicious IPs 15

Notes:

- **No sensitive data.** This is purely a simulation for learning.

Possible output:

```
Monitoring 4 suspicious IPs
ALERT: Packet 1 from 192.168.1.100
OK: Packet 2
ALERT: Packet 3 from 172.16.254.23
OK: Packet 4
```